

ICTS-ICTP Winter School on Quantitative Systems Biology 2025

From Data to Models

Discovering dynamical models from time series data

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Data
Time series data



Model
Dynamical model
Ordinary or stochastic
differential equations



- Where will such a tool be useful?
- Estimating dynamical models from data.
- A substitute/precursor to estimate coarse-grained dynamical descriptions from detailed agent-based/cellular automaton simulations. [2]
- Dimensionality reduction from high-dimensional models to lower-dimensional descriptions.

1. Jhavar, J. Noise-induced schooling of fish. *Nature Physics* 18.

2. Nabeel, A. , Jadhav, V. et al. Data-driven discovery of stochastic dynamical equations of collective motion. *Phys. Biol.* 20, 056003 (2023).

Data: time series data

Time series sampled at some discrete time points: $x_1, x_2, x_3, \dots, x_T$.

The time intervals need not necessarily be uniform,
but for this tutorial we will assume uniformly spaced samples, with a sampling interval Δt .

Model: Stochastic Differential Equations

$$\frac{dx}{dt} = f(x) + g(x) \cdot \eta(t)$$

Stochastic Differential Equations

$$\frac{dx}{dt} = \boxed{f(x)} + g(x) \cdot \eta(t)$$


Deterministic part

Characterizes the deterministic
or average dynamics

Stochastic Differential Equations

$$\frac{dx}{dt} = f(x) + g(x) \cdot \eta(t)$$



Stochastic part

Characterizes fluctuations around
the average dynamics

Stochastic Differential Equations

$$\frac{dx}{dt} = f(x) + g(x) \cdot \eta(t)$$



Fluctuations

$\mathcal{N}(0,1)$, independent at each time step.

Stochastic Differential Equations

$$\frac{dx}{dt} = f(x) + g(x) \cdot \eta(t)$$

↑
Strength of the
fluctuations.

Stochastic Differential Equations

$$\frac{dx}{dt} = f(x) + g(x) \cdot \eta(t)$$

↑
Strength of the
fluctuations.

State-dependent (multiplicative) noise.

Problem setting

Given time series data $x_1, x_2, \dots, x_T$ sampled with a sampling interval Δt , estimate a stochastic differential equation of the form

$$\frac{dx}{dt} = f(x) + g(x) \cdot \eta(t)$$

that describes the dynamics of x .

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that describes the dynamics of x .

In other words, estimate $f(x)$ and $g^2(x)$ from time series measurements of x .

Important: the goal is to discover interpretable *mathematical expressions* for f and g .

Estimating SDEs from data

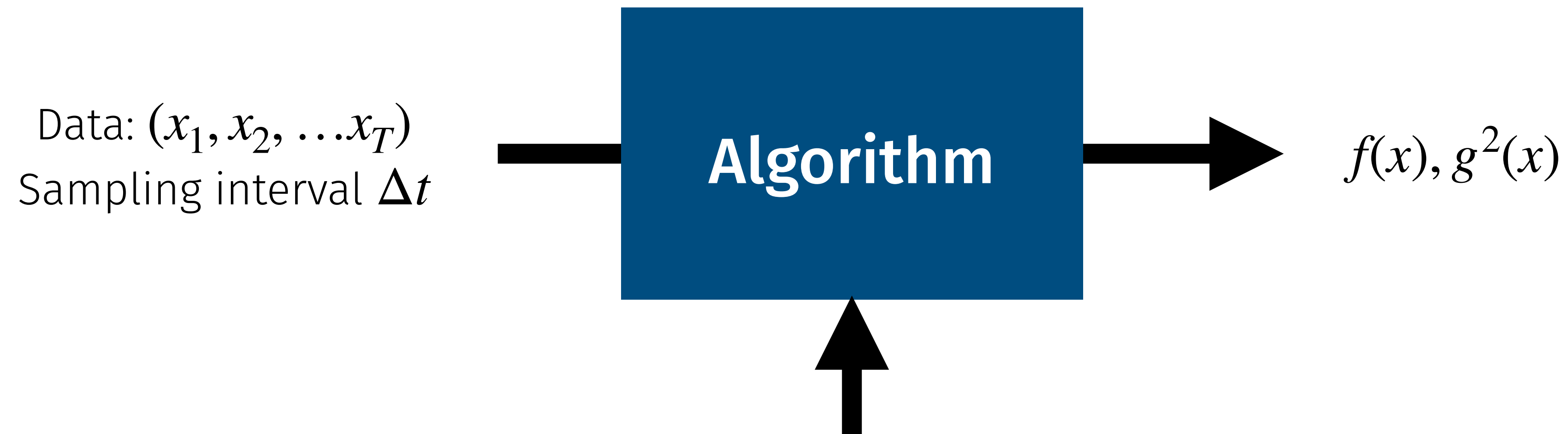
Key idea: Express f (or g^2) as a linear combination of ‘basis’ functions:

$$f(x) = \sum_{i=1}^K \alpha_i f_i(x)$$

The goal then is to estimate α_i ’s.

For a parsimonious model, we need a *sparse* linear combination: many α_i ’s can be 0.

Estimation algorithm as a black box



Need to specify:

The library of candidate functions: $f_1, f_2, \dots, f_K$

A sparsity parameter α

Estimating SDEs from data

The library of basis functions:

- Usually a good choice: polynomial terms $1, x, x^2, \dots$
 - $1, x, y, xy, x^2, y^2, \dots$ for 2-variable systems, etc.
 - The models we have seen so far in the workshop can be captured by this basis.

Estimating SDEs from data

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- Usually a good choice: polynomial terms $1, x, x^2, \dots$
 - $1, x, y, xy, x^2, y^2, \dots$ for 2-variable systems, etc.
 - The models we have seen so far in the workshop can be captured by this basis.
- Other basis sets can be chosen depending on the problem.
- Sometimes, symmetries or constraints inherent in the problem can help you choose the basis set smartly.

Estimating SDEs from data

- Too few terms in the library: the basis set may not be rich enough to capture the dynamics of the model.
 - E.g. The actual data has Allee effects, but the library has only terms upto degree 2.
- Too many terms in the library: possibility of overfitting.

Estimating SDEs from data

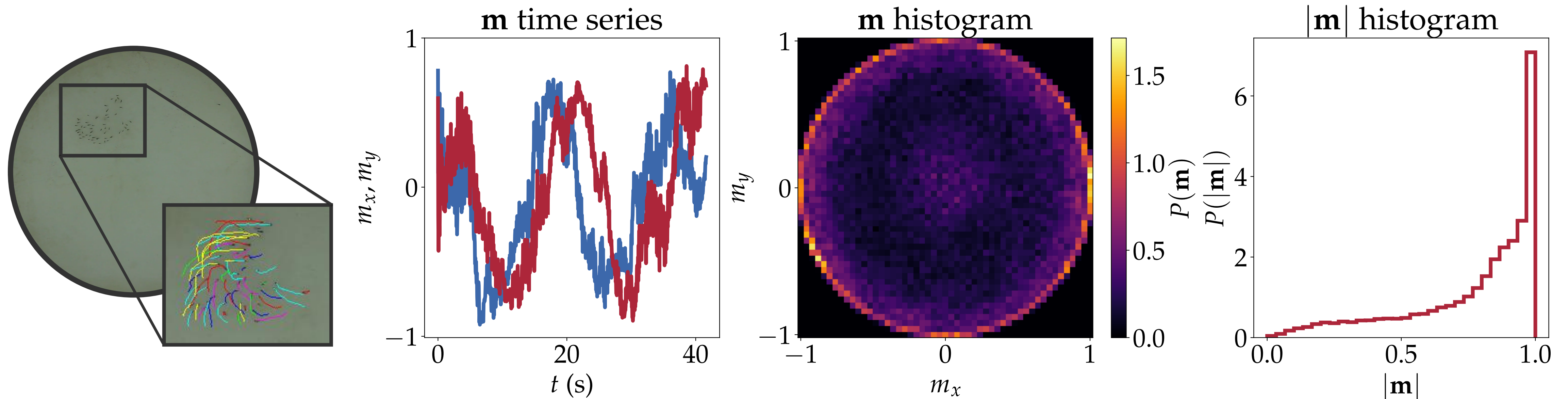
Sparsity parameter α determines how aggressively you prune terms.

- High α : Aggressive pruning, favour parsimony over fit accuracy.
- High α : Less aggressive pruning, favour fit accuracy over parsimony.

The library and the sparsity parameter decide the balance between *model parsimony* and *model fit accuracy*.

Example

Example: Modelling the stochastic dynamics of fish schools

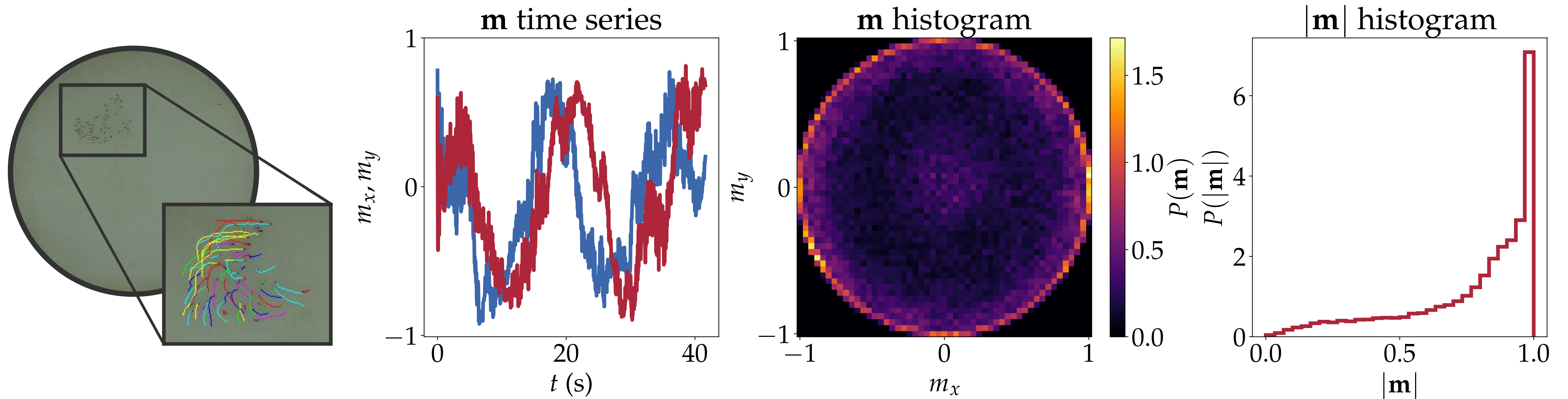


Jhawar, J. Noise-induced schooling of fish. Nature Physics 18.1.

Nabeel, A. et al. Discovering stochastic dynamical equations from ecological time series data. The American Naturalist 734083 (2024)

Example

Example: Modelling the stochastic dynamics of fish schools



$$f(\mathbf{m}) = -\alpha \mathbf{m}$$
$$g^2(\mathbf{m}) = \beta \left(1 - |\mathbf{m}|^2\right)$$

Example

Tutorial notebook: Dynamics of interacting species.

Details of the estimation algorithm

For a fixed x :

- What is $\left\langle \frac{dx}{dt} \right\rangle$?
- What is $\text{Var} \left(\frac{dx}{dt} \right)$?

Details of the estimation algorithm

1. Estimate the derivatives. $\delta x_i = \Delta x_i / \Delta t$ is an estimate of $\frac{dx}{dt}(t = t_i)$.
2. Find the functional form of $f(x)$ by fitting $f_j(x_i)$'s against δx_i .
3. Find the residual fluctuations, $r_i = \delta x_i - f(x_i)$.
4. Find the functional form of $g(x)$ by fitting $g_j(x_i)$ against r_i .

Validating the discovered model

How to know if the model produced by the algorithm is a good model for the dynamics?

Does the model produce dynamics with the same statistics as the data?
Compare histograms and autocorrelations.

Validating the discovered model

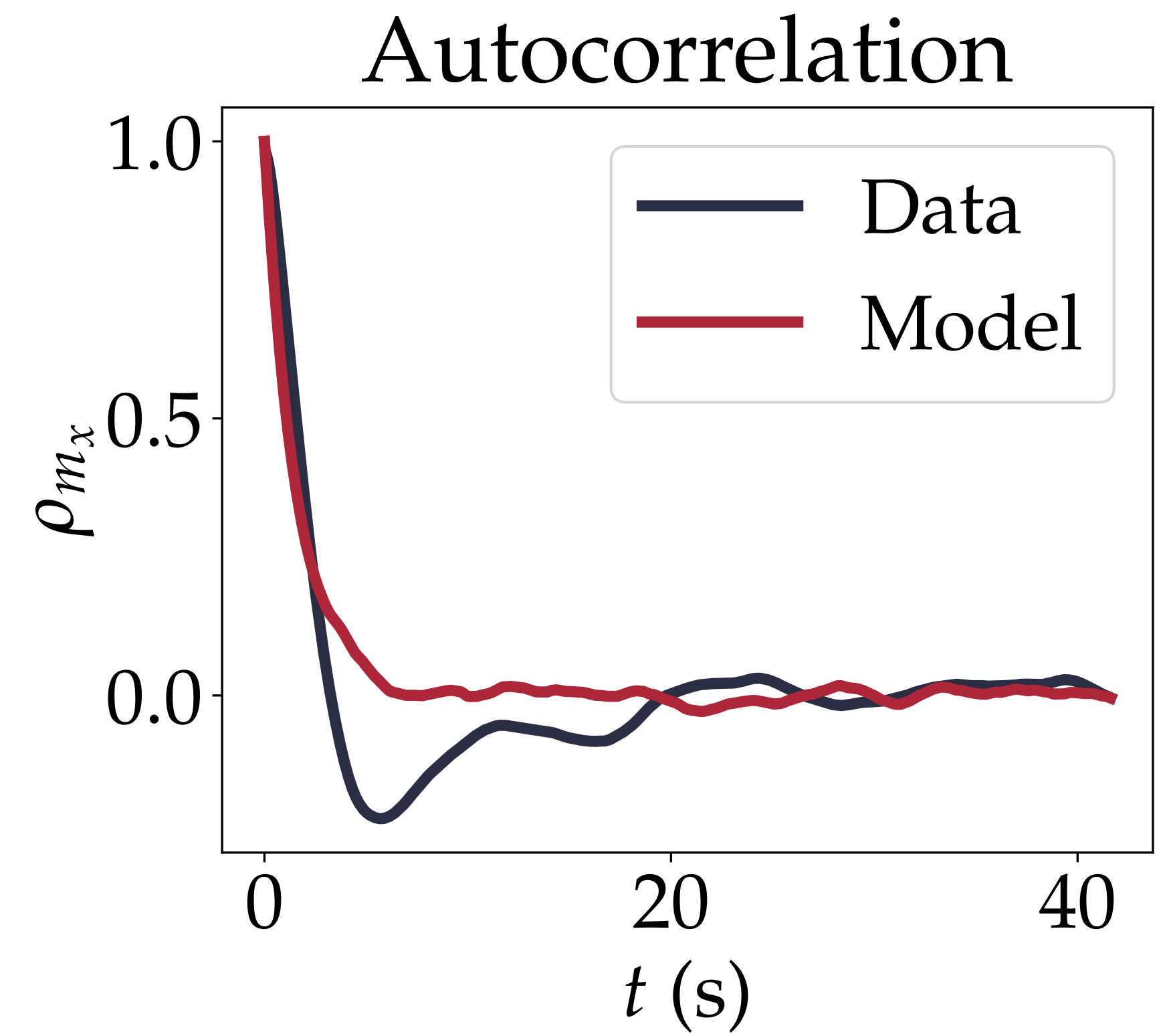
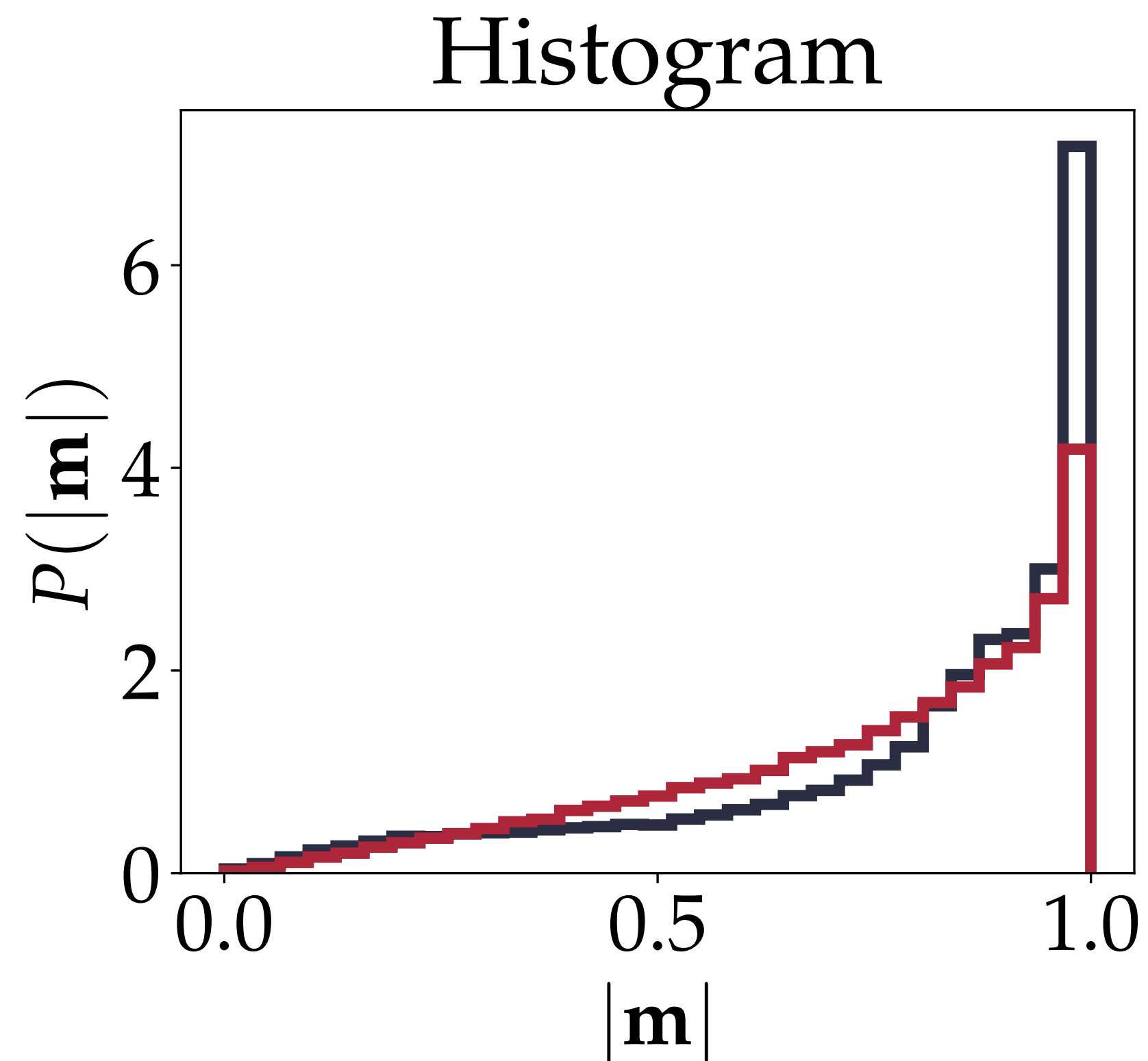
How to know if the model produced by the algorithm is a good model for the dynamics?

Are there residual dynamics in the model that are not captured by the model?

Look at the residual autocorrelation.

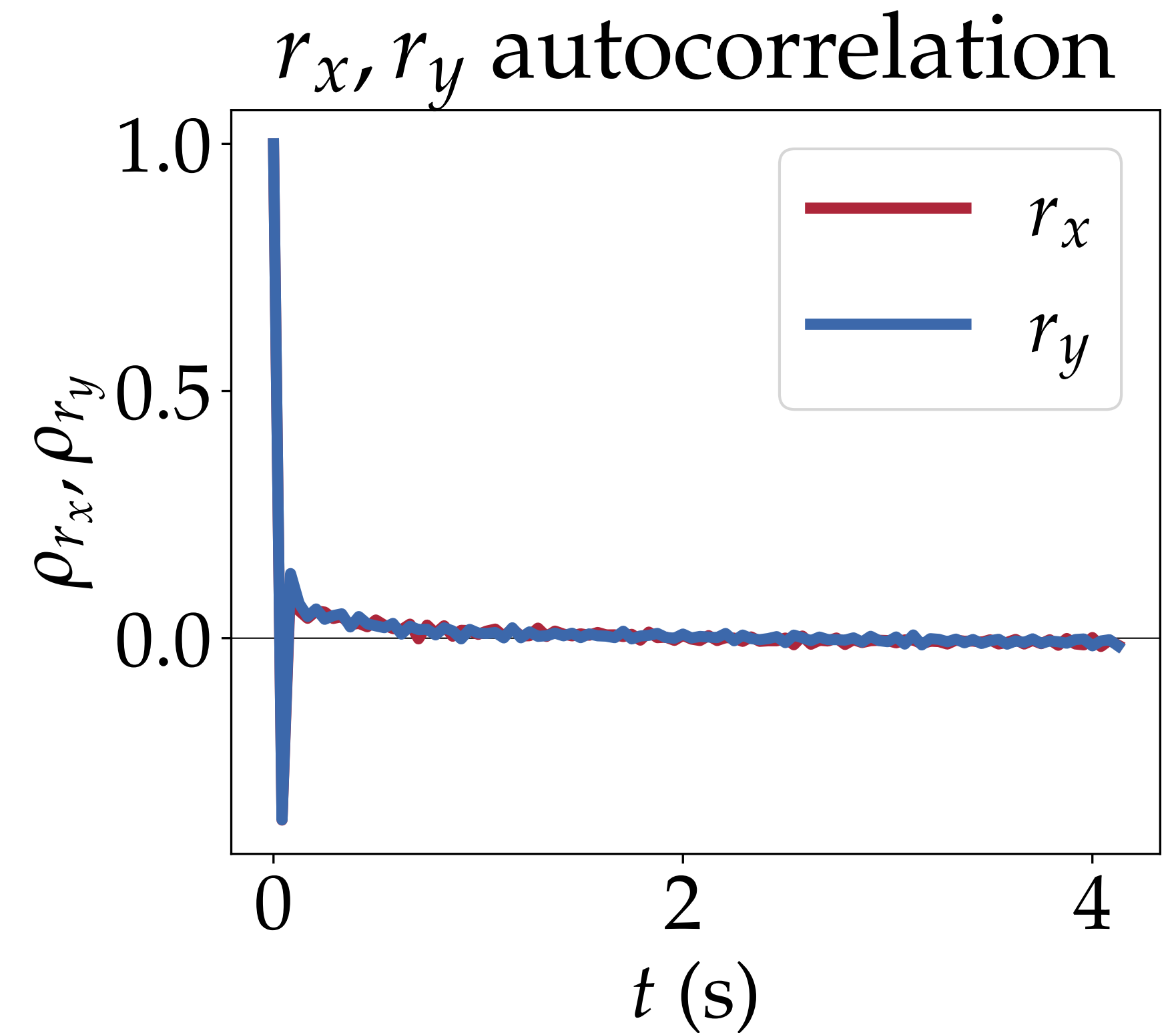
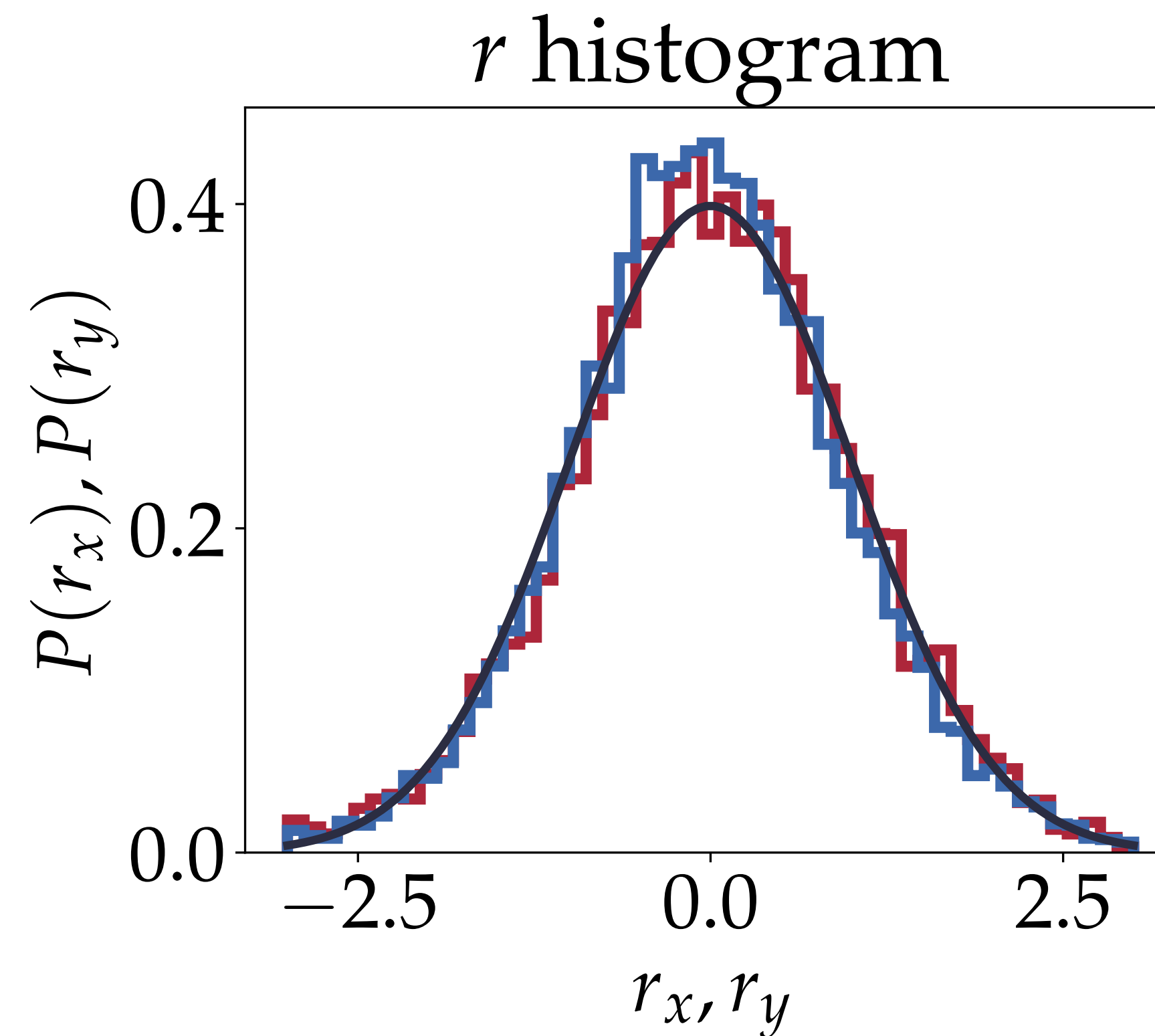
Validating the discovered model

Example: Modelling the stochastic dynamics of fish schools
Model versus data statistics



Validating the discovered model

Example: Modelling the stochastic dynamics of fish schools
Statistics of the residuals



Caveats

Sampling interval: The assumption is that the linear approximation $\frac{\Delta x_i}{\Delta t_i}$ is a good approximation.

- If Δt is too large, this may not hold true.

Higher dimensions: When the dimension of x increases, the number of terms you may need to consider may grow very quickly.

- Can be sometimes countered by considering symmetries or constraints in the system.
- How to choose (i) the library? (ii) the sparsity parameter?

Measurement noise: large measurement noise can complicate things.

References

Tutorial repo: **<https://shorturl.at/OR2VE>**

Python library (PyDaddy): **pydaddy.readthedocs.io**

- Nabeel, A. et al. Discovering stochastic dynamical equations from ecological time series data. The American Naturalist 734083 (2024)
- Nabeel, A. , Jadhav, V. et al. Data-driven discovery of stochastic dynamical equations of collective motion. Phys. Biol. 20, 056003 (2023)

References

Discovering ODEs from data:

- Brunton, S. L., Proctor, J. L. & Kutz, J. N. Discovering governing equations from data by sparse identification of nonlinear dynamical systems. Proc Natl Acad Sci USA 113, 3932–3937 (2016).

Python Library (PySINDy): **<https://pysindy.readthedocs.io/en/latest/>**

References

More rigorous approaches for discovering SDE models from data (takes into account measurement noise etc.):

- Frishman, A. & Ronceray, P. Learning Force Fields from Stochastic Trajectories. Phys. Rev. X 10, 021009 (2020).
- (Thesis) Ronceray, P. Learning dynamical models from stochastic trajectories. Preprint at <http://arxiv.org/abs/2406.02363> (2024).

References

Symbolic regression — allows more complex functions other than just linear combinations.

PySR: **<https://astroautomata.com/PySR/>**